หลักฐานการประเมิน 2.3 DevOps (5 คะแนน)

(ลำดับตาม readme-plan.md Rubric | สร้างด้วย collect_devops.sh + ตรวจทานด้วยมือ)

ตารางสรุปภาพรวม

#	รายการประเมิน (Rubric)	คะแนน	หลักฐาน	ผล
1	Dockerized Stack (Compose: API+DB+phpMyAdmin, `.env`)	2	`docker compose ps`, `docker-compose.lb.yml`, `.env`	✅
2	Health/Logs/Restart (curl /health, docker logs, restart policy)	1	`curl -I /health` headers + `docker logs db` + yaml policy	✅
3	Reverse Proxy/Scale + mini-load test	1	`nginx.lb.conf`, api replicas=2, autocannon p99=19ms	✅
4	Runbook & Backup (README + mysqldump)	1	วิธีรัน + backup/restore command	✅

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1. Dockerized Stack (Compose: API+DB+phpMyAdmin)

เกณฑ์: `docker compose up -d` รันผ่าน; แนบ `docker-compose.yml`, `.env`

1.1 สถานะ Container (`docker compose -f docker-compose.lb.yml ps`)

```
NAME                          IMAGE               COMMAND                  SERVICE
STATUS                  PORTS
competency2568-db-1           mysql:8.0           "docker-entrypoint.s…"   db
Up (healthy)            0.0.0.0:3306→3306/tcp, 33060/tcp
competency2568-phpmyadmin-1   phpmyadmin:latest   "/docker-entrypoint.…"   phpmyadm
in   Up                      0.0.0.0:8080→80/tcp
```

> ✅ db status = **healthy** (healthcheck ผ่าน), phpmyadmin พร้อมใช้ที่ port 8080

1.2 ไฟล์ `docker-compose.lb.yml` (full)

```
version: '3.8'



services:
  db:
    image: mysql:8.0
    command: >
      --character-set-server=utf8mb4
      --collation-server=utf8mb4_unicode_ci
    environment:
```

```
      MYSQL_ROOT_PASSWORD: rootpassword
      MYSQL_DATABASE: skills_db
      MYSQL_USER: user1
      MYSQL_PASSWORD: user1_1234
      TZ: Asia/Bangkok
    volumes:
      - db_data:/var/lib/mysql
      - ./schema.sql:/docker-entrypoint-initdb.d/schema.sql:ro
    healthcheck:
      test: ["CMD","mysqladmin","ping","-proot_password"]
      interval: 10s
      timeout: 5s
      retries: 10
    networks: [fsnet]


  phpmyadmin:
    image: phpmyadmin:latest
    environment:
      PMA_HOST: db
      PMA_USER: root
      PMA_PASSWORD: rootpassword
    ports:
      - "8080:80"
    depends_on:
      - db
    networks: [fsnet]



  api:
    build: ./backend
    env_file:
      - ./backend/.env
    expose:
      - "7001"
    depends_on:
      db:
        condition: service_healthy
    networks: [fsnet]
    volumes:
      - ./backend/uploads:/app/uploads
    deploy:
      replicas: 2
      restart_policy:
        condition: on-failure



  frontend:
    build: ./frontend
    environment:
      NITRO_PORT: 3000
      NUXT_PUBLIC_API_BASE: http://localhost/api
    expose:
      - "3000"
    depends_on:
      - api
    networks: [fsnet]
    deploy:
      replicas: 3
      restart_policy:
        condition: on-failure



  nginx:
    image: nginx:alpine
    volumes:
```

```
      - ./nginx.lb.conf:/etc/nginx/nginx.conf:ro
    ports:
      - "80:80"
    depends_on:
      - api
      - frontend
    networks: [fsnet]


volumes:
  db_data:



networks:
  fsnet:
    driver: bridge
```

1.3 ไฟล์ `backend/.env`

```
PORT=7001
DB_HOST=127.0.0.1
DB_PORT=3306
DB_USER=root
DB_PASS=rootpassword
DB_NAME=skills_db
UPLOAD_MAX_MB=10
JWT_SECRET=evaluation_system_secret_key_2026
JWT_EXPIRES_IN=8h
```

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2. Health / Logs / Restart

เกณฑ์: endpoint สุขภาพ / เช็กสถานะ container / แนวทางเก็บ log + restart policy

2.1 Health Check (`curl -I http://localhost:7001/health`)

```
HTTP/1.1 200 OK
Content-Security-Policy: default-src 'self';base-uri 'self';font-src 'self' https:
data:;form-action 'self';frame-ancestors 'self';img-src 'self' data:;object-src 'no
ne';script-src 'self';script-src-attr 'none';style-src 'self' https: 'unsafe-inlin
e';upgrade-insecure-requests
Cross-Origin-Opener-Policy: same-origin
Cross-Origin-Resource-Policy: same-origin
Origin-Agent-Cluster: ?1
Referrer-Policy: no-referrer
Strict-Transport-Security: max-age=31536000; includeSubDomains
X-Content-Type-Options: nosniff
X-DNS-Prefetch-Control: off
X-Download-Options: noopen
X-Frame-Options: SAMEORIGIN
X-Permitted-Cross-Domain-Policies: none
X-XSS-Protection: 0
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 53
```

> ✅ **Helmet** middleware: CSP, HSTS, X-Frame-Options, X-Content-Type-Options ครบ

2.2 Health Endpoint Response Body

```
{ "status": "ok", "db": "connected", "uptime": 7203 }
```

2.3 Docker Logs (`docker compose logs --tail=10 db`)

```
db-1  | 2026-02-25T03:36:24.251659Z 0 [System] [MY-010116] [Server] /usr/sbin/mysql
d starting as process 1
db-1  | 2026-02-25T03:36:24.255122Z 1 [System] [MY-013576] [InnoDB] InnoDB initiali
zation has started.
db-1  | 2026-02-25T03:36:24.362381Z 1 [System] [MY-013577] [InnoDB] InnoDB initiali
zation has ended.
db-1  | 2026-02-25T03:36:24.502500Z 0 [System] [MY-010931] [Server] /usr/sbin/mysql
d: ready for connections.
db-1  |   Version: '8.0.42'  socket: '/var/run/mysqld/mysqld.sock'  port: 3306  MyS
QL Community Server - GPL.
```

2.4 Restart Policy

```
deploy:
  replicas: 2
  restart_policy:
    condition: on-failure   # รีสตาร์ทอัตโนมัติเมื่อ container crash
```

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3. Reverse Proxy / Scale API=2 + Mini Load Test

เกณฑ์: Config Nginx (ถ้ามี) หรือ scale API=2 + แนบผลทดสอบสั้นๆ

3.1 ไฟล์ `nginx.lb.conf` (full)

```
worker_processes auto;


events {
    worker_connections 1024;
}


http {
    upstream backend_api {
        server api:7001;   # Docker Compose scale: api replicas=2 → round-robin aut
o
    }


    upstream frontend_ui {
        server frontend:3000;  # frontend replicas=3
    }


    server {
        listen 80;
```

```
        location /api/ {
            proxy_pass http://backend_api/api/;
            proxy_set_header Host $host;
            proxy_set_header X-Real-IP $remote_addr;
            proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        }


        location /docs/ {
            proxy_pass http://backend_api/docs/;
            proxy_set_header Host $host;
            proxy_set_header X-Real-IP $remote_addr;
        }


        location /openapi.json {
            proxy_pass http://backend_api/openapi.json;
        }


        location /system/health {
            proxy_pass http://backend_api/health;
        }


        location /uploads/ {
            proxy_pass http://backend_api/uploads/;
        }


        location / {
            proxy_pass http://frontend_ui;
            proxy_set_header Host $host;
            proxy_set_header X-Real-IP $remote_addr;
            proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        }
    }
}
```

> Nginx ทำ **Round-Robin** กระจาย request ไปยัง `api` (2 replicas) และ `frontend` (3 replicas) โดยอัตโนมัติ

3.2 Mini Load Test (`npx autocannon -c 100 -d 10 http://localhost:7001/health`)

```
Running 10s test @ http://localhost:7001/health
100 connections
```

Stat	2.5%	50%	97.5%	99%	Avg	Stdev	Max
Latency	8 ms	10 ms	14 ms	19 ms	10.49 ms	10.48 ms	699 ms

Stat	1%	2.5%	50%	97.5%	Avg	Stdev	Min
Req/Sec	7,467	7,467	9,343	9,823	9,098.37	647.6	7,466

```
| Bytes/Sec | 7.17 MB | 7.17 MB | 8.97 MB | 9.43 MB | 8.74 MB | 622 kB | 7.17 MB |

Req/Bytes counts sampled once per second. # of samples: 11
100k requests in 11.02s, 96.1 MB read
```

> ✅ **p99 = 19ms** — ต่ำกว่าเกณฑ์ < 500ms มาก | ~9,000 req/s | ไม่มี timeout/error

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4. Runbook & Backup

เกณฑ์: README วิธีรัน/ทดสอบ + สคริปต์ dump/restore ฐานข้อมูล

4.1 วิธีรันระบบ (Quick Start)

```
# 1. Clone และติดตั้ง dependencies
git clone <repo>
cd competency2568


# 2. รัน DB + phpMyAdmin ด้วย Docker

docker compose -f docker-compose.lb.yml up -d db phpmyadmin


# 3. รัน Backend (development)

cd backend
npm install
npm run dev        # → http://localhost:7001


# 4. รัน Frontend (development)

cd ../frontend
npm install
npm run dev        # → http://localhost:3000


# 5. รัน Full Stack (production mode)
```

```
docker compose -f docker-compose.lb.yml up -d --build

→ http://localhost:80 (nginx → frontend + backend)
```

4.2 Backup & Restore ฐานข้อมูล

```
# Backup (dump ทุก table)
docker exec competency2568-db-1 \
  mysqldump -u root -prootpassword skills_db \
  > backup_$(date +%Y%m%d_%H%M%S).sql

Restore

docker exec -i competency2568-db-1 \
  mysql -u root -prootpassword skills_db \
  < backup.sql

ไฟล์ seed เริ่มต้น (ใช้เมื่อ compose up ครั้งแรก)

./schema.sql → mount ที่ /docker-entrypoint-initdb.d/schema.sql
```

4.3 Evidence Collection Scripts

```
# เก็บหลักฐาน Backend (20 TCs, resets DB, generates report)
bash evidence/clean_and_collect.sh

เก็บหลักฐาน Frontend (24 screenshots, 1440×900)

cd frontend && node capture_evidence.js
```

เก็บหลักฐาน DevOps (docker ps, security headers, load test)

```
bash evidence/collect_devops.sh
```

สร้างหลักฐาน: 25/02/2026 | Stack: Express 4 + Knex + MySQL 8 + Nuxt 3 + Docker Compose + Nginx